

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
B.Tech Programme in ELECTRONICS ENGINEERING (A batch)
Second Internal Examination, July 2024
Second semester (2023 scheme)
23-200-0201 LINEAR ALGEBRA & TRANSFORM TECHNIQUES

Time : 2 hours

Max.Marks : 30

Course Outcome (CO)

On completion of this course the student will be able to

- CO1. Solve linear system of equations and to determine Eigen values and vectors of a matrix
CO2. Exemplify the concept of vector space and sub space.
CO3. Determine Fourier series expansion of functions and transform.
CO4. Solve linear differential equation and integral equation using Laplace transform.

PART A

(Answer all questions)

Max.marks

1. Find the ^{inverse} Laplace transform of $\log \left(\frac{s(s+1)}{s^2+1} \right)$ 2 L1 (CO4)(PO1)
2. Prove that $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$ 2 L1 (CO4)(PO1)
- ✓ 3. Find the Laplace transform of $\frac{(\cos at - \cos bt)}{t}$ 2 L2 (CO4)(PO1)
4. Find the fourier cosine transform of $f(x) = \begin{cases} x, & 0 < x < 1 \\ 2-x, & 1 < x < 2 \\ 0, & x > 2 \end{cases}$ 2 L1 (CO3)(PO1)
- ✓ 5. Find the Laplace transform of $(t e^{-2t} \sin 3t)$ 2 L2 (CO4)(PO1)

PART B

(Answer any two questions)

6. a) Apply convolution theorem to evaluate $L^{-1} \frac{s^2}{(s^2+a^2)(s^2+b^2)}$

- b) Solve the following equation by method of transform $y'' + 4y' + 3y = e^{-t}$
 $y(0) = y'(0) = 1$

10 L3 (CO4)(PO1)

OR

- 7.a) Using unit step function, find the Laplace transform of $f(t) = \begin{cases} \cos t, 0 \leq t < \pi \\ \cos 2t, \pi \leq t < 2\pi \\ \cos 3t, t \geq 2\pi \end{cases}$

- b) Find the Laplace transform of the function

$$f(t) = \begin{cases} \sin \omega t, & 0 < t < \pi/\omega \\ 0, & \pi/\omega < t < 2\pi/\omega \end{cases}$$

10 L3 (CO4)(PO1)

- 8 a) Find the Fourier series expansion of the function $f(x) = x \cdot x^2$ from $x = -\pi$ to π

- b) Express $f(x) = x$ as a half range cosine series in $(0, \pi)$ and hence deduce the sum of series $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$

10 L1 (CO3)(PO1)

OR

- 9 a). Find the complex form of Fourier series of $f(x) = e^{-x}$ in the interval $-1 < x < 1$

- b). Find the Fourier transform of $f(x) = \begin{cases} 1, & |x| < 1 \\ 0, & |x| > 1 \end{cases}$ Hence evaluate $\int_0^{\infty} \frac{\sin x}{x} dx$

10 L1 (CO3)(PO1)